

```
#practice tasks
import numpy as np
a=np.array(42)
b=np.array([1,2,3,4,5])
c=np.array([[1,2,3,4,5],[6,7,8,9,10]])
print("\n")

print(a.ndim)
print(b.ndim)
print(c.ndim)
print("\n")

array=[[1,2,3],[4,5,6]]
m=np.matrix(array)
print(m)
print("\n")

M=np.matrix('1 2;3 4')
print(M)
print("\n")

A=np.array([[1,2,3],[4,5,6]])
B=np.array([[1,2],[1,1],[1,0]])
C=A.dot(B)
print(C)
print("\n")

d=np.array([[1,2],[3,4],[5,6]])
print(d.transpose())
print("\n")

A = np.array([2, 4, 6, 8, 10])
print("A[0] =", A[0]) # First element
print("A[2] =", A[2]) # Third element
print("A[-1] =", A[-1]) # Last element
```

```
0
1
2

[[1 2 3]
 [4 5 6]]

[[1 2]
 [3 4]]

[[ 6  4]
 [15 13]]

[[1 3 5]
 [2 4 6]]
```

```
#task1
import numpy as np

arr = np.array([[1.2, 0.0, -3.4],
                [0.0, 5.6, 7.8]])

print("Original array:")
print(arr)
print(arr.dtype)
```

```
bool_arr = arr.astype(bool)

print("\nConverted to bool:")
print(bool_arr)
print(bool_arr.dtype)
```

```
Original array:
[[ 1.2  0. -3.4]
 [ 0.   5.6  7.8]]
float64
```

```
Converted to bool:
[[ True False  True]
 [False  True  True]]
bool
```

```
#task2
import numpy as np

arr = np.array([10, 20, 30, 40, 50])

print("Original array:")
print(arr)

updated_arr = np.insert(arr, 2, 55)

print("Updated array:")
print(updated_arr)
```

```
Original array:
[10 20 30 40 50]
Updated array:
[10 20 55 30 40 50]
```

```
#task3
import numpy as np

sequence = np.arange(0, 101, 2)

print(sequence)
```

```
[ 0  2  4  6  8 10 12 14 16 18 20 22 24 26 28 30 32 34
 36 38 40 42 44 46 48 50 52 54 56 58 60 62 64 66 68 70
 72 74 76 78 80 82 84 86 88 90 92 94 96 98 100]
```

```
#task4
import numpy as np

A = np.array([[1, 2],
              [3, 4]])

B = np.array([[5, 6],
              [7, 8]])

result = np.matmul(A, B)

print("Matrix multiplication result:")
print(result)
```

```
Matrix multiplication result:
[[19 22]
 [43 50]]
```

```
#task5
import numpy as np

arr = np.array([5, 10, 15, 20, 25])

number = 15

if number in arr:
    print("Number is present")
else:
    print("Number is not present")
```

Number is present